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Machine Learning Approaches for Analyzing and Predicting Concurrent Delays in Construction Projects

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Abstract

Construction projects are highly complex and often vulnerable to delays due to multiple overlapping factors. This study employs machine learning techniques to analyse and predict concurrent delays, a critical issue in the construction industry. A synthetic dataset was generated to represent key factors influencing delays, including labour hours, resource availability, weather conditions, and project complexity. Preprocessing steps such as normalization and outlier handling were applied to ensure data quality. K-means clustering was used to identify patterns in delay occurrences, while a Random Forest classifier provided predictive insights. The model demonstrated robust performance, achieving high accuracy, precision, and recall in delay prediction. Comparative analysis with traditional methods highlighted the advantages of machine learning in identifying complex relationships between variables. This study provides actionable insights for stakeholders to mitigate delays and optimize project outcomes using data-driven approaches.

Keywords: Delay prediction, Machine learning, K-means clustering, Random Forest, Construction projects, Project optimization

1 Introduction

The construction industry plays a pivotal role in driving economic growth, as it is closely linked to infrastructure development, urbanization, and societal progress [1]. Despite its significance, the industry faces numerous challenges, with project delays being one of the most persistent and impactful issues [2]. Delays in construction projects can have severe consequences, not only in terms of cost overruns but also in relation to the quality of the final product, stakeholder satisfaction, and the overall reputation of contractors and project managers [3]. Among the various types of delays, concurrent delays, which occur when two or more delay events overlap, are particularly problematic [4]. These delays are complex and difficult to manage because the simultaneous occurrence of multiple factors often complicates their underlying causes, leading to challenges in accountability, responsibility assignment, and effective resolution [5].

Concurrent delays can stem from a wide array of interrelated factors that complicate their identification and quantification. Resource shortages, such as the unavailability of materials, equipment, or skilled labour, often interact with external factors like adverse weather conditions, regulatory changes, or communication inefficiencies [6]. Additionally, unforeseen circumstances such as accidents, labour strikes, and supply chain disruptions can further exacerbate the situation [7]. The interdependencies between these variables complicate the process of determining how each factor individually impacts the overall project timeline, making it difficult to adopt a straightforward approach for analysing and addressing delays [8].

Traditional methods of delay analysis, such as the Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT), have been widely used in the construction industry for decades [9]. While these techniques provide valuable insights into project scheduling and milestone tracking, they are often limited by their reliance on deterministic models and manual analysis. CPM and PERT approaches work under the assumption of a relatively stable environment, where delays are generally predictable and can be mitigated by adjusting individual tasks . However, these methods fall short when dealing with the dynamic, multi-variable conditions of large-scale construction projects, where factors interact in complex, non-linear ways. Their scalability and ability to capture hidden patterns and interdependencies are limited, especially when projects involve vast amounts of data, multiple stakeholders, and complicated project schedules[8].

With the rapid advancement of machine learning (ML) technologies, new opportunities have emerged to enhance delay analysis in construction projects . ML techniques, particularly supervised learning and unsupervised learning, allow for the processing of large, complex datasets, making it possible to uncover patterns and relationships that may not be readily apparent through traditional analysis methods. In the context of construction delays, machine learning enables the identification of factors influencing delays, the prediction of delay occurrences, and the assessment of their potential impact on project outcomes [7].

Among the various machine learning methods, K-means clustering and Random Forest classification stand out as effective techniques for delay analysis [9]. K-means clustering, an unsupervised learning algorithm, is well-suited to identifying patterns in delay occurrences by grouping similar delay events together. This enables the discovery of distinct clusters of delays, which can then be analysed to understand the underlying causes and characteristics of those delays. In contrast, Random Forest, a powerful ensemble classification model, can be used to predict the likelihood of delays by analysing the interdependent variables that contribute to disruptions. Through this approach, project managers can gain insights into how different factors such as labour hours, weather conditions, resource availability, and project complexity interact and influence the likelihood of delays occurring [2].

The main advantage of using machine learning in this context is its ability to analyse large datasets with multiple variables, allowing for more accurate and scalable predictions than traditional methods. Machine learning models can also evolve over time as new data becomes available, leading to continual improvements in delay prediction accuracy. Moreover, these models empower project managers with the ability to make proactive, data-driven decisions, such as adjusting schedules, reallocating resources, or implementing contingency plans, before delays escalate into more significant problems [9].

This study focuses on the application of machine learning to analyse and predict concurrent delays in construction projects. To simulate real-world scenarios, a synthetic dataset was created, encompassing key factors influencing delays, such as labour hours, resource availability, weather conditions, and project complexity [1]. The dataset was pre-processed using several techniques to ensure data quality and consistency, such as normalization to scale the data and outlier handling to remove any anomalous data points. Following this, K-means clustering was applied to group delay occurrences based on their similarity, while Random Forest classification was used to predict the likelihood of delays based on the identified patterns. The results of these analyses demonstrated the ability of machine learning models to accurately identify and predict delays, with high accuracy, precision, and recall [3].

Comparative analysis between the results from machine learning models and traditional methods highlighted the superior ability of machine learning to handle complex and large datasets, providing insights that traditional methods could not easily uncover. By using machine learning, construction stakeholders can move from reactive to proactive management, anticipating delays and taking corrective actions before they have a significant impact on project timelines [9].

2 Literature Review

Amur Ali Mussallam Al Amri et al. proposed that delays in construction projects significantly impact timelines, finances, stakeholder relationships, and overall success. These delays can lead to cost overruns, extended durations, and strained relationships. Understanding their causes and effects is crucial for improving project management and ensuring timely, successful project completion [2].

Hudoyo et al. highlighted that delays in construction are complex, stemming from factors like resource shortages, adverse weather, regulatory barriers, and unforeseen disruptions. These factors hinder project progress, causing delays, increased costs, and challenges in maintaining quality and stakeholder satisfaction. Addressing these causes is key to improving project outcomes[6].

O et al. emphasized that delays affect budgets, schedules, and quality, often leading to rushed work, client dissatisfaction, and reputational damage for contractors and project managers. They also highlighted that, unlike traditional methods, ML can analyze vast data, uncover hidden patterns, and deliver accurate, scalable predictions[3].

Fatch et al. argued that concurrent delays, when multiple delays occur simultaneously, create a complex challenge by overlapping factors that complicate responsibility allocation, resource management, and finding solutions. This strains schedules and budgets, hindering successful project completion. They also explored ML's role in addressing concurrent delays, emphasizing its advantages over traditional methods and its potential to revolutionize project management practices [4].

Alemu et al. highlighted that concurrent delays complicate accountability and responsibility allocation, increasing uncertainty and making resolution more complex and time-consuming. They also noted that the rise of machine learning (ML) in recent years offers a promising alternative for overcoming these challenges in construction project management[5].

Osman et al. identified delays as one of the most critical challenges in the construction industry, consistently affecting project timelines, costs, and success. They also noted that ML techniques have significant potential in identifying the causes of delays and predicting their likelihood and impact on project timelines[3].

Rahman et al. noted that traditional methods like the Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT), though widely used, have limitations in managing the non-linear and dynamic nature of modern construction projects [9].

Wang et al. referred to traditional methods like CPM and PERT, which rely on deterministic models assuming stable, predictable conditions. These assumptions rarely match the dynamic and uncertain nature of real-world construction projects [10].

Ibrahim et al. pointed out that traditional methods, like CPM and PERT, struggle to capture complex interdependencies and fail to provide actionable insights in large-scale projects with overlapping delays [8].

Roumeissa et al. emphasized the growing need for advanced, data-driven approaches to address the limitations of traditional methods in construction project management [1].

The effects of construction project delays on budgets, schedules, and stakeholder relationships have been thoroughly examined in previous research. The limits of conventional techniques like CPM and PERT in tackling these issues are acknowledged in previous research, which also links delays to several circumstances, such as resource scarcity, weather, and regulatory obstacles. By highlighting its capacity to examine sizable datasets, reveal hidden patterns, and enhance predictive accuracy, machine learning (ML) is examined in several works as a viable substitute. Nevertheless, existing research mostly concentrates on concurrent delays or general delays causes without completely incorporating machine learning-based solutions designed for dynamic, large-scale building contexts. Furthermore, little research has been done on the systematic modeling of overlapping delays and interdependencies using machine learning (ML) to provide project managers with useful insights. Through the development of a sophisticated machine learning framework, our study fills these gaps by predicting delays and offering a methodical, data-driven strategy for reducing their effects, improving project planning, and allocating resources as efficiently as possible.

3 Methodology

The use of machine learning techniques has become crucial in data analysis because they provide strong instruments for classification, pattern identification, and decision-making. The goal of this approach is to use preprocessing, clustering, classification, and synthetic data synthesis to create a predictive model. The procedure uses Random Forest classification for predictive analysis and K-means clustering for data segmentation. Effective data handling and model optimization are guaranteed by this structured pipeline, which enhances performance.

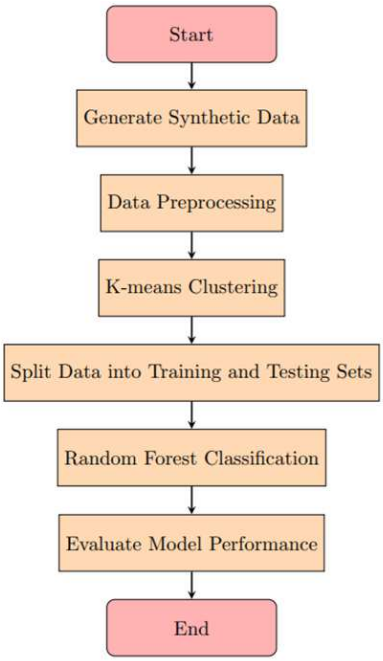


Fig. 1 End-to-End Machine Learning Workflow: Synthetic Data Generation, Preprocessing, K-Means Clustering, and Random Forest Classification for Performance Evaluation

The study begins by defining its objectives and outlining the workflow, which includes identifying the problem statement, research questions, and expected outcomes. Clear goals ensure a focused and effective methodology. A well-structured plan integrates data generation, preprocessing, clustering, classification, and evaluation seamlessly, aligning tasks with objectives for accurate results and insights.

In the first step, synthetic data is generated to simulate real-world scenarios when actual datasets are unavailable or insufficient. This involves defining the data structure, including the number of features, data types, and distributions. Tools such as Python libraries, including NumPy and Scikit-learn are utilized to create random samples or simulated datasets. Generating synthetic data allows for controlled experimentation, enabling the study to test models under specific conditions while maintaining data variability and relevance.

Next, data preprocessing is carried out to clean and transform the raw data. This includes handling missing values, removing duplicates, normalizing features, and encoding categorical variables. Scaling numerical features standardizes the data, improving the performance of machine learning algorithms. Outlier detection and feature selection further enhance the dataset's quality. Proper preprocessing minimizes computational overhead, reduces potential biases, and ensures the data is consistent, error-free, and ready for analysis.

The pre-processed data is then subjected to K-means clustering, an unsupervised learning technique that partitions data into distinct groups based on shared characteristics. This algorithm identifies patterns and relationships in the data by assigning data points to the nearest cluster centre. The optimal number of clusters was determined using methods such as the elbow method and the silhouette score. K-means clustering uncovers hidden structures, providing insights into data distribution and guiding subsequent steps like data splitting and classification.

After clustering, the dataset is divided into training and testing subsets. Typically, 70-80% of the data is allocated for training, with the remaining 20-30% reserved for testing. This random splitting ensures both subsets represent the overall data distribution, avoiding selection bias. Proper partitioning prevents overfitting and enables a

reliable evaluation of the model's generalization capability, which is critical for robust machine learning.

A Random Forest classifier is then trained on the training dataset for predictive analysis. This ensemble learning method builds multiple decision trees and combines their outputs to improve accuracy and mitigate overfitting. During training, hyperparameters such as the number of trees and the maximum depth are optimized to improve the model's performance. The classifier learns patterns in the data and predicts outcomes for new instances effectively. Random Forest also provides feature importance scores, making it a versatile tool for classification tasks.

The trained model's performance is evaluated using the testing dataset. Key metrics such as accuracy, precision, recall, F1-score, and the confusion matrix are computed to assess the model's predictive capability. Evaluation highlights the model's strengths and limitations, identifying areas for improvement. Out-of-Bag (OOB) error estimation ensures that the results are reliable and not affected by random variations. This thorough evaluation validates the model's robustness, generalizability, and alignment with the study's objectives.

Finally, the methodology concludes with a summary of findings, interpretation of results, and insights. The outcomes of model evaluation are analyzed to determine if the objectives were achieved. The entire process is documented, highlighting successes, challenges, and areas for future improvement. If the results are satisfactory, recommendations and applications will be proposed; otherwise, further research or methodological adjustments will be suggested.

4 Data Set Preparation

The preparation of the dataset is a crucial step to ensure the quality, validity, and reliability of subsequent analysis. In this study, synthetic data was generated to simulate a variety of factors that influence construction project delays. These factors include labor hours, resource availability, weather conditions, project complexity, and several other relevant features. The dataset consists of 1,000 samples, each represented by eight features that collectively capture the multifaceted nature of construction projects.

To model the occurrence of concurrent delays effectively, labels were assigned to the dataset based on a predefined threshold condition applied to specific feature combinations. For instance, if labor hours, weather conditions, and project complexity exceeded a particular value collectively, the label for concurrent delays was set to 1. Otherwise, the label was set to 0. This binary labelling ensures a clear distinction between projects with and without concurrent delays, facilitating supervised learning.

Table 1 Factors Affecting Construction Projects

Factor	**Description**
Labor Hours	Total hours worked on-site, affecting productivity and delays.
Resource Availability	Availability of essential materials and equipment.
Weather Conditions	Impact of adverse weather (e.g., rain, snow) on work.
Project Complexity	Measured by task count and dependencies.
Communication Efficiency	Effectiveness of team communication in coordination.
Site Conditions	Physical site factors (e.g., terrain, accessibility).
Regulatory Delays	Delays due to permits, inspections, or legal issues.
Shift Patterns	Work shifts (e.g., day/night) influencing labor availability.

Dependent Variable: Occurrence of Concurrent Delays The dependent variable is a binary label representing whether concurrent delays occur in a project. A value of 0 indicates no concurrent delays, while a value of 1 signifies their presence. This binary classification facilitates the analysis of contributing factors and supports the application of predictive modelling techniques like Random Forest or Logistic Regression.

Data Normalization and Cleaning

Before applying clustering and classification techniques, the dataset was pre-processed to ensure consistency and quality:

Normalization: All features were scaled to a uniform range, eliminating the effects of magnitude differences across variables. This step enhances the performance of machine learning algorithms, especially those sensitive to feature scaling.

Outlier Handling: Outliers and anomalies were identified using statistical techniques such as Z-scores or IQR (Interquartile Range) and addressed appropriately. This prevents extreme values from skewing the analysis.

Generalizability: Care was taken to design synthetic data that reflects real-world scenarios, ensuring the findings are applicable to practical construction project management.

This well-prepared dataset forms the foundation for reliable clustering and classification techniques, offering insights into the factors contributing to construction project delays and enabling the development of predictive models to mitigate them.

5 Model Training and Results

Our machine learning model was developed and assessed using a standardized training pipeline. To manage missing values, eliminate noise, and guarantee consistency, the dataset was initially pre-processed. After that, we used K-means clustering to divide the data into segments, which improved model performance by identifying innate patterns.

After then, the dataset was divided into training and testing sets to guarantee that the classes were distributed evenly. A typical allocation of data was 20% for testing, which ensures an objective assessment of performance, and 80% for training, which enables the model to learn from a variety of patterns.

Because of its resilience and capacity to manage high-dimensional data, we chose Random Forest Classification as our predictive model. To improve model performance, hyperparameter adjustment was done, and cross-validation was used to reduce overfitting.

Finally, to provide a thorough assessment of the model's efficacy, its performance was evaluated using common evaluation metrics like accuracy, precision, recall, and F1-score.

K-Means Clustering of Construction Data

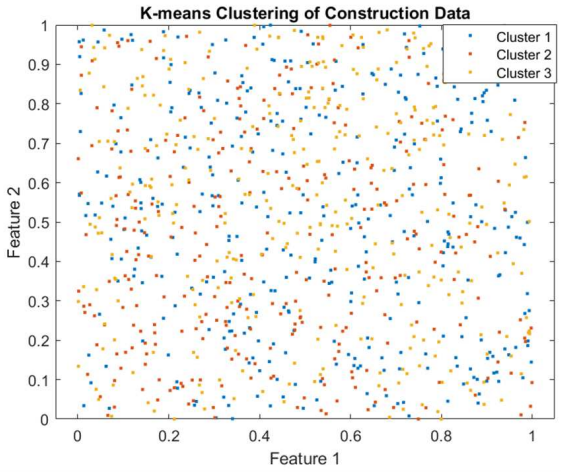


Fig. 2 Application of K-Means Clustering on a Construction Dataset: Grouping Data Points into Distinct Clusters Based on Material Properties, Workforce Productivity, and Project Stages.

The figure 2 demonstrates the application of K-means clustering to a construction-related dataset, grouping data points into three distinct clusters: Cluster 1 (blue), Cluster 2 (red), and Cluster 3 (yellow). This scatterplot effectively captures relationships and patterns within the dataset, with axes labeled as Feature 1 (x-axis) and Feature 2 (y-axis), representing the underlying variables.

K-means is an unsupervised machine learning algorithm that partitions data into a predefined number of clusters by minimizing intra-cluster variance and maximizing inter-cluster separation. Each data point is assigned to the nearest cluster center, which iteratively adjusts its position until an optimal configuration is achieved. The resulting clusters reveal meaningful groupings within the data, such as material properties, workforce productivity categories, or construction project stages. This visualization highlights the algorithm's ability to identify distinct patterns, providing a foundation for resource allocation, risk assessment, and performance evaluation.

Cluster Centers with Classified Data

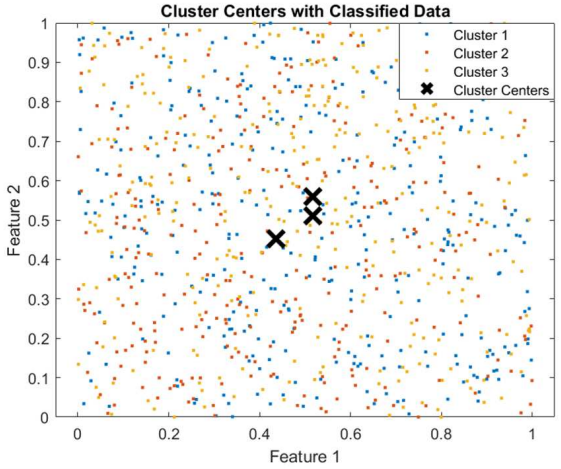


Fig. 3 Enhanced K-Means Clustering Visualization with Cluster Centroids: Incorporating Bold Black Crosses to Represent Centroid Locations and Illustrating Geometric Relationships Between Clusters and Their Centers in a Construction Dataset.

Building on the clustering results, Fig 3 incorporates the cluster centers, represented by bold black crosses, into the scatterplot. These centroids, calculated as the mean positions of all points within their respective clusters, define the boundaries of each cluster. The dataset remains classified into the same three clusters using the K-means algorithm, with iterative refinements of cluster memberships and centroid locations.

The inclusion of centroids enhances the interpretability of the results by illustrating geometric relationships between clusters and their centers. The proximity of data points to centroids reflects strong associations with specific clusters, while well-spaced centroids indicate clear distinctions between groups. This visualization is valuable for segmenting construction resources, identifying operational characteristics, or categorizing performance levels. By examining the centroids, stakeholders can gain deeper insights into the dataset's structure and make informed decisions. Centroid positions can also inform downstream analyses, such as cohesion evaluation, outlier detection, and feature impact exploration.

Random Forest Classification Performance

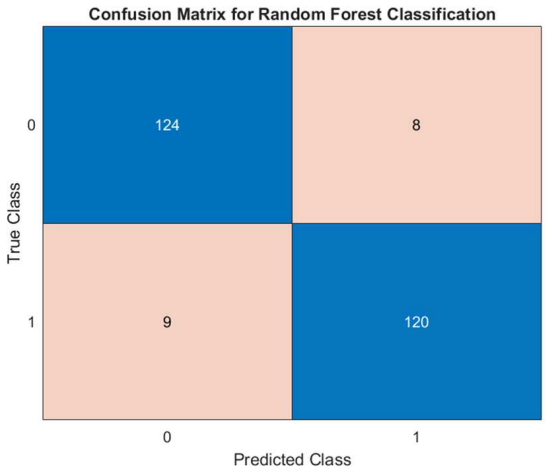


Fig. 4 Confusion Matrix for Random Forest Classification: Evaluating Model Performance on a Binary Classification Problem with 124 True Negatives, 120 True Positives, 8 False Positives, and 9 False Negatives.

The confusion matrix evaluates a Random Forest classification model applied to a binary classification problem, comparing true labels with model predictions. Key metrics include 124 true negatives (correctly predicted as Class 0), 120 true positives (correctly predicted as Class 1), 8 false positives (Class 0 misclassified as Class 1), and 9 false negatives (Class 1 misclassified as Class 0). This results in 244 correctly classified instances and 17 misclassifications, showcasing strong performance.

Random Forest's ensemble learning approach, which combines multiple decision trees, enhances prediction accuracy and reduces overfitting. However, misclassifications warrant attention. False positives may lead to unnecessary actions, such as inspecting non-defective areas, while false negatives could result in overlooked defects, potentially causing significant issues. These findings highlight opportunities for optimization, such as hyperparameter tuning, feature selection, or addressing data imbalances. Despite some errors, the confusion matrix underscores the model's potential for tasks requiring precise and reliable predictions.

Random Forest OOB Error Analysis

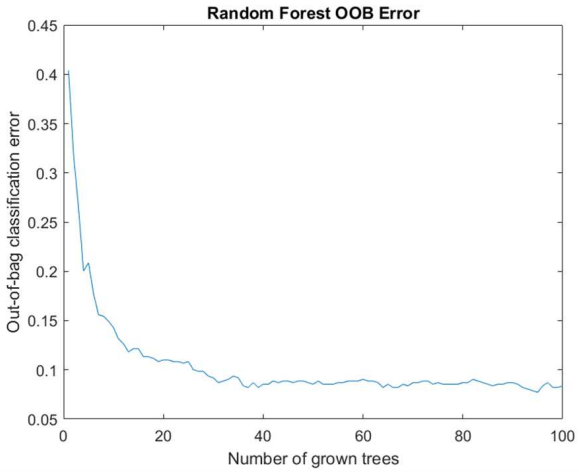


Fig. 5 Random Forest OOB Error vs. Number of Trees: Analyzing the Relationship Between Tree Count and Out-of-Bag Classification Error, Demonstrating Improved Generalization and Stability with Increasing Trees.

Fig 5 illustrates the relationship between the number of trees in a Random Forest model and the Out-of-Bag (OOB) classification error. The X-axis shows the number of grown trees (0 to 100), while the Y-axis represents the OOB classification error (0.05 to 0.45). Initially, with few trees, the error is high (0.4) due to limited predictive power. As more trees are added, the error decreases sharply, reflecting improved generalization and stability. Beyond 30 trees, the reduction in error diminishes, stabilizing at 0.1 with around 100 trees, indicating optimal performance.

The sharp initial error decrease demonstrates the value of ensemble learning in reducing overfitting and improving accuracy. The curve's flattening beyond a certain point highlight diminishing returns, where additional trees offer minimal gains. OOB error, calculated via bootstrap sampling, provides a reliable performance measure without requiring a separate validation set. This graph emphasizes the balance between computational efficiency and accuracy, suggesting that 50–100 trees achieve optimal results for this dataset, while maintaining the robustness and generalization characteristic of Random Forests.

6 Conclusion

This study demonstrated the effectiveness of machine learning techniques in analyzing and predicting concurrent delays in construction projects. By leveraging K-means clustering and Random Forest classification, the research identified critical factors influencing delays and provided a data-driven approach to mitigating their impact. The results showed that machine learning models outperformed traditional methods, offering greater accuracy, scalability, and insight into complex relationships between variables.

The findings emphasize the potential of machine learning to transform delay management in construction by enabling proactive decision-making, optimizing resource allocation, and improving project scheduling. With continued advancements in data collection and model refinement, these techniques can further enhance project efficiency and reduce risks associated with delays. Future research could explore the integration of real-world construction data and the application of deep learning models for even more precise delay predictions.

By adopting machine learning-based delay prediction systems, stakeholders in the construction industry can shift from reactive problem-solving to a more strategic, preventive approach, ultimately ensuring more successful project outcomes.

Conflict-of-Interest Statement

The authors declare that they have no conflicts of interest regarding this study.

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